B. Subsequence Update

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: constructive algorithms, data structures, greedy, sortings | https://codeforces.com/problemset/problem/2063/B

After Little John borrowed expansion screws from auntie a few hundred times, eventually she decided to come and take back the unused ones.But as they are a crucial part of home design, Little John decides to hide some in the most unreachable places — under the eco-friendly wood veneers.

You are given an integer sequence $a_1, a_2, \ldots, a_n$, and a segment $[l,r]$ ($1 \le l \le r \le n$).

You must perform the following operation on the sequence **exactly once**.

- Choose any **subsequence**[] of the sequence a, and reverse it. Note that the subsequence does not have to be contiguous.

Formally, choose any number of indices $i_1,i_2,\ldots,i_k$ such that $1 \le i_1$ lt i_2 lt … lt $i_k \le n$. Then, change the i_x-th element to the original value of the i_{k-x+1}-th element simultaneously for all $1 \le x \le k$.

Find the **minimum value** of $a_l+a_{l+1}+\ldots+a_{r-1}+a_r$ after performing the operation.

[]A sequence b is a subsequence of a sequence a if b can be obtained from a by the deletion of several (possibly, zero or all) element from arbitrary positions.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, l, r ($1 \le l \le r \le n \le 10^5$) — the length of a, and the segment $[l,r]$.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($1 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case, output the minimum value of $a_l+a_{l+1}+\ldots+a_{r-1}+a_r$ on a separate line.

Example

Input

```
6
2 1 1
2 1
3 2 3
1 2 3
3 1 3
3 1 2
4 2 3
1 2 2 2
5 2 5
3 3 2 3 5
6 1 3
3 6 6 4 3 2
```

Output

```
1
3
6
3
11
8
```

Note

On the second test case, the array is a=[1,2,3] and the segment is [2,3].

After choosing the subsequence a_1,a_3 and reversing it, the sequence becomes [3,2,1]. Then, the sum a_2+a_3 becomes 3. It can be shown that the minimum possible value of the sum is 3.